Information Processing and Management 62 (2025) 104164



Contents lists available at ScienceDirect

Information Processing and Management

journal homepage: www.elsevier.com/locate/ipm



Knowledge enhanced representation learning network for drug recommendation

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ARTICLE INFO

Keywords:
Drug recommendation
Representation learning
Class imbalanced distribution

ABSTRACT

Drug recommendation systems have attracted considerable attention within medical healthcare, which aim to deliver personalized and efficacious drug combinations tailored to patients' clinical records. Through extensive investigation, we identify two key issues with existing methods: (1) class imbalance distribution, where common diseases occur more frequently than rare ones, resulting in biased and insufficient patient representations; and (2) inadequate modelling of historical medications, where the historical drugs may contain drug information that is valuable for current medical treatment is often overlooked. In this paper, we propose a Knowledge Enhanced Representation Learning (KERL) network for drug recommendation. To address the first issue, we introduce external medical knowledge, using disease entities of different granularities to enhance patient representation. Meanwhile, we construct multiple medical knowledge graphs based on extracted entities and design a graph knowledge enhancement mechanism to integrate global clinical information, alleviating the imbalanced distribution of medical entities. To address the second issue, we design a dual-path drug representation network to model longitudinal historical information from both visit-level and drug-level perspectives. Extensive experiments on two real-world datasets MIMIC-III and MIMIC-IV demonstrate the effectiveness of the proposed KERL in drug recommendation task. Specifically, our KERL achieves improvements of 2.15%, 2.09%, 1.92% and 2.09%, 2.74%, 1.96% over current state-of-the-art methods in terms of F1-score, PRAUC, and Jaccard, respectively.

1. Introduction

Addressing the management of patients presenting with intricate multimorbidity has been acknowledged as a challenging issue, owing to the complexities inherent in disease and medication interactions, as well as the problematic potential for adverse drug reactions (Zhang et al., 2017). Clinicians typically formulate medication prescriptions based on the patient's health status and mix personal intuition and experience. However, due to a lack of knowledge or experience, many recommended drug combinations may be suboptimal (Yu et al., 2024). In the United States, the cost of adverse drug interactions and drug events exceeds $136 billion annually (Lazarou et al., 1998). Recently, the increasing abundance of healthcare data, e.g. Electronic Health Records (EHRs),

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https://doi.org/10.1016/j.ipm.2025.104164
Received 26 October 2024; Received in revised form 29 January 2025; Accepted 24 March 2025
Available online 18 April 2025
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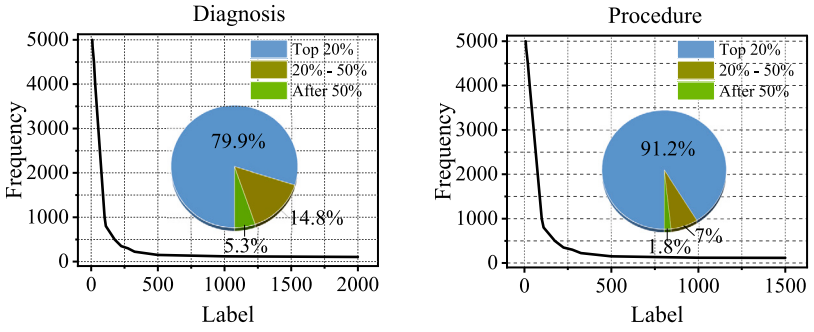


Fig. 1. Statistical analysis of the distribution of medical entities (i.e., diagnosis and procedure codes) on benchmark datasets, reflecting the extreme imbalance and long-tail phenomenon.

enables researchers to develop advanced predictive models for clinical decision-making assistance (Bhoi et al., 2021; Tan et al., 2022). Drug recommendation aims to provide doctors with candidate drug set based on the patient's health condition, represented by diagnosis and procedure codes, which is crucial for improving prescription efficiency (Sun et al., 2022; Wang, Chen, Pi, Yue, et al., 2021; Zheng et al., 2021).

Drug recommendation involves mapping medical entity sequences (diagnosis and procedure sequences) into vector space to encode the patient's health status (Yang, Xiao, Glass, & Sun, 2021; Zhang et al., 2023). The existing methods are primarily divided into instance-based and longitudinal-based recommendation methods. The instance-based approaches aim to recommend medication based on the patient's current medical visit without considering past medical history (Wu et al., 2024). LEAP (Zhang et al., 2017) employs a recurrent encoder to capture the dependency relationships between medical entities during the current visit and uses a content-based attention mechanism to extract crucial disease-drug mappings. SMR (Gong et al., 2021) constructs heterogeneous graphs for safe medicine recommendations by sharing comprehensive low-dimensional representations of patients, drugs, and diseases. On the other hand, longitudinal-based approaches specifically focus on extracting information related to the current visit from the longitudinal data of patients to enrich their representation. For instance, DGCL (Li, Zhang, Li, et al., 2023) designs a distance detection loss to calculate the similarity between current and past cases, thereby significantly enhancing recommendation performance. Furthermore, MIRNet (Li, Zhang, Hou, et al., 2023) develops a multi-visit filter to extract critical historical visit links and reduce unnecessary noise during the entire process. Additionally, many studies have also emphasized incorporating diverse medical knowledge sources, such as drug substructures (Yang, Xiao, Ma, et al., 2021; Yang et al., 2023), EHR graphs (Shi et al., 2024; Wu, Qiu, et al., 2022), symptom data (Sun et al., 2022; Tan et al., 2022), and medication dosage titrations (Bhoi et al., 2024), to enhance feature representation. Although previous models have shown promising performance, there are still many limitations, as follows:

- **Class imbalanced problem.** In EHR systems, patients with complicated diseases are often associated with numerous medical codes (Ji et al., 2024). As shown in Fig. 1, the distribution of diagnosis and procedure entities in the code sequence is highly imbalanced, presenting a long-tail phenomenon. The top 20% of the two codes account for 79.9% and 91.2% of this code's total occurrence frequency, respectively, and 80% of the code's occurrence frequency is less than a quarter of the total occurrence frequency of all codes. Relying solely on data-driven methods makes the model focus more on densely distributed medical information, resulting in biased and insufficient patient representations (PRs). The patient representation is the basis for accurate drug recommendations. Therefore, it is crucial to construct a comprehensive and unbiased patient representation.
- **Inadequate historical drug modelling.** The existing longitudinal-based methods typically compare the similarity between past visits and current patient representations, integrating historical visit information into a global patient representation based on different weights. These advanced deep learning techniques utilize multi-layer neural networks to capture the long-term visit dependencies among multiple visits. However, an increase in the number of layers can result in heightened mutual information between the derived PR and the input patient data, accompanied by the potential loss of crucial information indicative of the patient's disease status and may introduce a lot of noise, affecting drug prediction (Wu et al., 2023). Meanwhile, they suffer from a single perspective to predict drugs based on the patient's health status, ignoring the modelling of historical medications for one patient. For specific patients (such as those with chronic diseases), directly selecting essential drugs from past medications is more effective and convenient for formulating current prescriptions.

To address the above challenges, we propose a Knowledge Enhanced Representation Learning (KERL) framework for drug recommendation. First, we design a knowledge enhanced patient representation module to integrate Clinical Classification Software (CCS) code knowledge. By jointly optimizing the prediction results on International Classification of Disease (ICD) and CCS code branches, the drug recommendation model can improve its ability to recognize low-frequency disease codes while maintaining sensitivity to common disease codes, thereby enhancing patient representation. Specifically, the diagnosis and procedure codes in EHRs are represented by ICD codes. As shown in Fig. 2, we introduce CCS codes for knowledge distillation, which can convert multiple ICD codes into coarse-grained CCS codes in many-to-one relationships. For example, in the ICD coding system, '427.31'

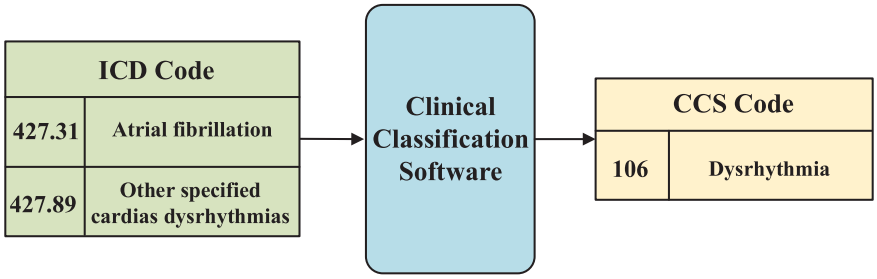


Fig. 2. Mapping a sample from multiple ICD codes to corresponding CCS codes by Clinical Classification Software for drug recommendation, which can be seen as a knowledge distillation process.

and '427.89' respectively represent 'Atrial fibrillation' and 'Other specified cardias dysrhythmias', which can be classified as a major category in the CCS system, namely 'Dysrhythmia', and can be represented by CCS code '106'. We extend the previous input by representing patients through two different types of codes and then formulate drug recommendations. In this way, these low frequency codes are merged into broader categories, thereby increasing data density and alleviating data sparsity of low-frequency codes. Moreover, we develop a dual-path drug representation network, modelling both at the visit level and the drug level, thus overcoming the limitations of previous approaches that focus only on patient representation at the visit level while ignoring historical drug modelling.

In summary, the contributions of KERL are as follows:

- We propose a drug recommendation model, KERL, which incorporates external medical knowledge to enrich patient representation and leverages historical drug information to promote more accurate recommendations.
- We introduce CCS code knowledge and construct multiple medical knowledge graphs to improve representation learning, which integrate disease codes of different granularities to alleviate bias caused by imbalanced medical entity distribution.
- We design a dual-path drug representation network to model drug prescriptions from both the visit and drug levels, which can extract patient longitudinal past drug information to select reusable drugs.
- Our method achieves SOTA performance on multiple benchmark datasets, i.e. MIMIC-III and MIMIC-IV datasets. Also, we provide comprehensive experiment analyses for the proposed approach.

2. Related work

2.1. Representation learning for drug recommendation

Representation learning aims to encode necessary information about entities, sequences and others into low-dimensional vector representation spaces (Hoang et al., 2024). The drug recommendation task involves patient representation learning and drug representation learning. The former refers to encoding the patient's diagnosis and procedure sequences for patient representation, each sequence containing multiple disease codes. The latter obtains drug representation by directly embedding drug sequences into the representation space or converting drugs into SMILES strings. Drugs are also indirectly characterized by mining the relationships between their internal molecules or functional groups. Representation learning in drug recommendation is vital to capture latent features and relationships, thereby enhancing the accuracy and efficiency of recommendation systems.

Specifically, for each code sequence, GAMENet (Shang, Xiao, et al., 2019), SafeDrug (Yang, Xiao, Ma, et al., 2021), and VITA (Kim et al., 2024) embed each code in the sequence to obtain an initialized vector representation and then concatenate the diagnosis and procedure sequences to obtain the patient representation. However, this patient representation is coarse-grained, and the interactions and long-term dependencies between medical codes in the sequence are ignored. Therefore, COGNet (Wu, Qiu, et al., 2022) introduces a multi-head self-attention mechanism to capture the functional relationships of medical codes within the same sequence. Moreover, for drug representation, the treatment of diseases by drug molecules to a certain extent depends on the properties of drug substructures. SafeDrug (Yang, Xiao, Ma, et al., 2021) and MoleRec (Yang et al., 2023) model drug substructures to fully utilize the perception information of drug substructures to enhance drug representation and recommendation safety. Although previous models have explored multiple methodologies for patient representation and drug representation, the long-tailed distribution of medical codes has not received attention.

2.2. Longitudinal-based drug recommendation methods

In drug recommendation systems, longitudinal-based approaches (Meng et al., 2024; Shang, Ma, et al., 2019; Wang, Chen, Pi & Yue, 2021; Wang et al., 2018; Wu, Qian, et al., 2022) capitalize on the inherent temporal dynamics within patients' extended medical histories to anticipate future prescriptions. DRMP (Ren et al., 2022) designs a temporal association learning mechanism to construct

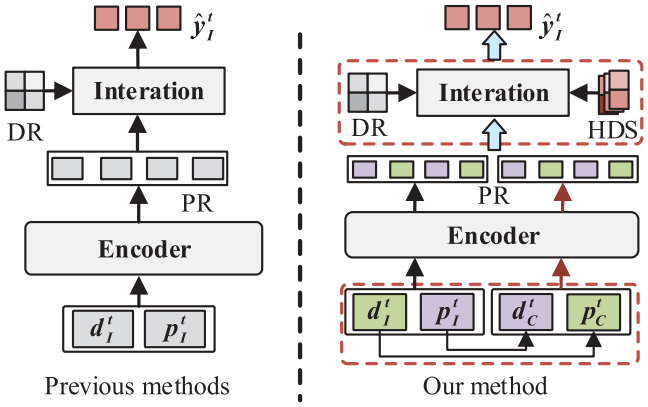


Fig. 3. An illustration of previous training methods and our proposed method. "PR" and "DR" denote patient representation and drug representation, respectively. The red line represents a new branch of knowledge distillation, termed the CCS code branch. We convert the diagnosis and procedure sequences, d_I^t and p_I^t, of ICD codes for the tth visit into corresponding CCS code sequences, d_C^t and p_C^t, and then embed them into a vector space. Our method utilizes the knowledge distillation technique to effectively characterize patients and mine reusable drugs from Historical Drug Sets (HDS).

co-occurrence graphs of medical items between near visits to capture information relationships at the visit level. Carmen (Chen et al., 2023) and MoleRec (Yang et al., 2023) use RNNs to separately model the temporal dependencies of diagnosis and procedure sequences across different visits. VITA (Kim et al., 2024), DrugRec (Sun et al., 2022), and ACDNet (Mi et al., 2024) utilize attention mechanisms to integrate valuable information from previous visits to obtain comprehensive patient representations. However, it is difficult for RNNs to accurately identify the most critical part of past medical information, and attention mechanisms allocate different weights by calculating the distance between multiple visits, which may easily introduce unnecessary noise.

Differently, MIRCON and COGNet, from a novel perspective, directly model patients' historical drug information. However, MIRCON only identifies valuable drugs from a recent visit, which makes modelling the patient's history of medication incomplete and insufficient. COGNet designs a hierarchical selection mechanism to comprehensively replicate useful drugs from all history records for one patient, achieving improved performance. However, this replication operation is coarse-grained, causing the model to predict redundant drugs, which enhances model performance but ignores safety considerations, resulting in a high DDI rate. These methodologies harness the temporal dependencies across time-stamped medical records, enabling predictive models to formulate personalized medication recommendations for impending healthcare needs.

2.3. Class imbalanced problem

The issue of class imbalance stems from the unequal distribution of labels within datasets (Chawla et al., 2004), often resulting in a head-tail pattern, where high-frequency labels dominate the "head" and low-frequency ones form the "tail". This problem is prevalent in most multi-label datasets (Tahir et al., 2012). A typical strategy to address it involves dividing classes into majority groups (frequent labels) and minority groups (infrequent labels) (Charte et al., 2015), followed by sampling techniques (Chawla et al., 2002; Kotsiantis & Pintelas, 2003) to either reduce the majority or augment the minority. However, modifying medical datasets is both resource-intensive and inefficient. Therefore, many automatic drug recommendation approaches are proposed to address this issue. For example, DrugRec (Sun et al., 2022) devises a causal graph for drug recommendation, addressing recommendation bias as a confounding factor and achieving debiased recommendations by exploring the causal effect. RAREMed (Zhao et al., 2024) designs a pretrain-finetune paradigm to improve prediction performance for rare diseases. Although the above methods successfully identify class imbalance distribution issues, they either neglect the integration of domain knowledge or demonstrate limited recommendation performance.

2.4. KERL in relation to exiting methods: similarities and differences

Unlike other deep learning models, KERL proposes a novel knowledge enhanced representation learning network that incorporates external domain knowledge to enhance representation learning and comprehensively improve drug recommendation performance.

In Fig. 3, we summarize the differences between our method and previous methods, mainly reflected in two aspects: integrating domain knowledge to alleviate the long tail distribution of medical entities to enhance representation learning and integrating historical information from a new perspective, where modelling the patient's history of medication.

One of the significant differences is that KERL effectively handles the imbalanced distribution of medical entities in drug recommendation. This imbalance is often overlooked by previous models, which only use a purely data-driven approach to learn entity representations. As a result, they embed diagnosis and procedure entities into representations and perform simple concatenation to obtain patient representations. However, this approach leads to biased and insufficient patient representations. In

Table 1
Notation for KERL.

Notation	Definition
$\mathcal{V}$	Patient health records
V^t	The tth visit for one patient
d, p, m	Diagnosis, procedure and medication set
E_{I_d}, E_{I_p}	Initialization embedding of diagnosis and procedure in ICD code branch
E_{C_d}, E_{C_d}	Initialization embedding of diagnosis and procedure in CCS code branch
v_I^t, v_C^t	Patient representation for two branches
$\mathcal{G}_d$, $\mathcal{G}_p$	Co-occurrence graph for two branches
G_d^e, G_p^e	Node embedding for $\mathcal{G}_d$, $\mathcal{G}_p$
M_e	Drug representation
$\hat{y}$	Drug usage probability

contrast, KERL introduces CCS knowledge and uses knowledge distillation to extract highly skewed medical codes into a larger code space, constructing multiple prediction branches and sharing information between different code prediction branches. Meanwhile, it constructs multiple medical knowledge graphs based on extracted entities and employ graph convolutional network to integrate global clinical information. This unique approach allows KERL to effectively alleviate the problem of imbalanced code distribution, enriching patient representation.

Moreover, KERL novelly designs the dual-path drug prediction module to predict drugs from different perspectives at the visit level and drug level. Previous models typically derive the ultimate patient representation by modelling patients' longitudinal historical visit data and subsequently interacting with drug representations to generate recommended drug combinations. However, these models overlook the direct exploration of reusable drugs within the historical drug sets, and inadequate patient representation may hinder the precise capture of valuable drug information, thereby introducing extraneous noise.

3. Preliminary

Definition 1 (*Patient Health Records*)**.** Electronic health records (EHRs) contain a large amount of medical information from patients during their visits, and the sequential visit records for each patient x can be denoted as $\mathcal{V}=[V_x^1, \cdots, V_x^{t-1}, V_x^t]$, where V_x^{t-1} represents the $(t-1)$-th visit for patient x. To simplify the expression, we omit the patient subscript x in the following text and the patient's visits are represented as $\mathcal{V}=[V^1, \cdots, V^{t-1}, V^t]$. Each visit consists of three medical code sequences: diagnosis sequence d, procedure sequence p and drug sequence m, where can be denoted as multi-hot vectors. The tth visit can be denoted as $V^t=\{d^t \in \{0,1\}^{|D|}, p^t \in \{0,1\}^{|P|}, m^t \in \{0,1\}^{|M|}\}$, where $|D|$, $|P|$ and $|M|$ are the total number of diagnosis, procedure and drug labels.

Definition 2 (*Knowledge Graph Construction*)**.** In each visit, we convert the diagnosis and procedure sequences (represented as ICD codes) into the CCS codes, respectively. Then, we count the co-occurrence frequency of diagnosis or procedure codes for different branches of all visits in EHRs. For example, for diagnostic codes, when ICD diagnosis code I_d and CCS diagnosis code C_d appear in one visit simultaneously, the co-occurrence frequency is increased by one. Finally, we obtain the diagnosis co-occurrence knowledge graph $\mathcal{G}_d=(N_{I_d}, N_{C_d}, \varepsilon_d)$ and procedure co-occurrence knowledge graph $\mathcal{G}_p=(N_{I_p}, N_{C_p}, \varepsilon_p)$ between different code branches, where N and ε denote node set and the set of edges, and each edge is accompanied by a weight representing the degree of association between codes of the same type in different branches. I and C represent ICD and CCS code branches, respectively.

Definition 3 (*Drug Recommendation*)**.** Construct a comprehensive representation of the patient's health status by combining past visit information $[V^1, \cdots, V^{t-1}]$ and current medical information (diagnosis sequence d^t and procedure sequence p^t), and then recommend the medication combination for current visit. In our framework, we convert the diagnosis and procedure ICD codes into the CCS codes to utilize the two code branches to predict drugs. Specifically, for tth visit of one patient, we leverage the diagnosis sequence d_I^t, procedure sequence p_I^t in ICD code branch and the diagnosis sequence d_C^t, procedure sequence p_C^t in CCS code branch to learn patient presentation, respectively. The patient presentation interacts with the drug representation to obtain the final set of drug predictions, respectively. In the end, we jointly train the drug prediction results of the two branches to obtain the final drug prediction combination. Necessary notations used in this paper are listed in the Table 1.

4. Method

4.1. Framework overview

In this section, we provide a detailed description with our proposed framework. Fig. 4 illustrates the overview of KERL, which consists of data input module, knowledge enhanced patient representation module and dual-path drug representation module:

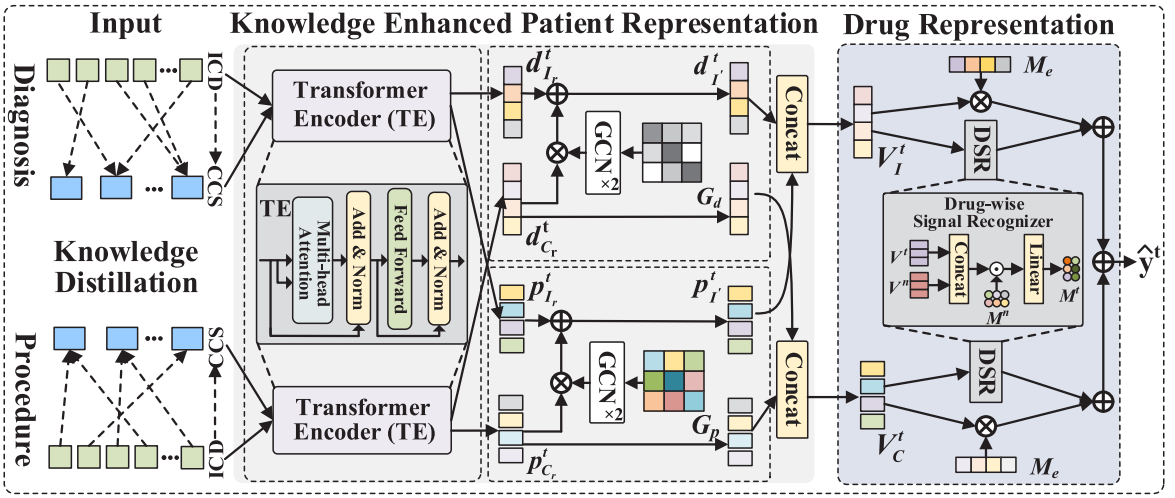


Fig. 4. The model overview of KERL, which consists of three modules: Input, Knowledge Enhanced Patient Representation, and Drug Representation. TE denotes the Transformer Encoder and DSR represents Drug-wise Signal Recognizer. V^t and V^n represent the tth visit and previous $t-1$ visits, respectively. $\otimes$, $\oplus$ and $\odot$ represent matrix multiplication, matrix addition and the dot product operation, respectively.

- **Input data.** As shown in Fig. 4, the first part uses patient clinical data as input (representing diagnosis and procedure codes). Next, we integrate clinical classification software to distil knowledge from ICD codes representing diseases, mapping fine-grained imbalanced ICD codes to a broader CCS code space to construct predictive branches for two types of codes: the ICD-drug branch and the CCS-drug branch.
- **Knowledge enhanced patient representation.** As shown in Fig. 4, the second part aims to learn comprehensive and unbiased patient representations. Specifically, it first utilizes the Transformer Encoder to model each sequence representation, capturing the long-term dependency of different medical codes. Then, it constructs multiple medical knowledge graphs, as the diagnosis co-occurrence knowledge graph and procedure co-occurrence knowledge graph are incorporated to capture the global relationships of different medical codes at the label level. Finally, it concats the final diagnosis and procedure representations of each code branch to obtain patient representations for different branches.
- **Dual-path drug representation.** As shown in Fig. 4, the last part focuses on how to obtain the accurate drug representation. It designs the dual-path drug prediction module to predict drugs from different perspectives at the visit level and drug level. The visit level aims to perform a interaction between patient representation and drug representation to predict new drugs. In the drug level, it leverages a Drug-wise Signal Recognizer (DSR) to guide the model in selecting drugs closely related to the current visit at each past visit. By adaptive aggregation, we obtain the final drug prediction combinations.

Each part of KERL is described sequentially below.

4.2. Data input

The imbalanced diagnosis and procedure code distribution in EHRs makes the model pay attention to occupying the majority of medical entities, resulting in biased patient representation. Therefore, we introduce CCS code knowledge for knowledge distillation and convert fine-grained ICD code sequences into coarse-grained CCS code sequences, alleviating patient representation bias caused by imbalanced information distribution.

First, we convert the diagnosis and procedure sequences of ICD d_I^t, p_I^t into CCS code sequences d_C^t, p_C^t and then embed them into a vector space to obtain corresponding embedding representations. Specifically, the health status of patients is encoded through diagnostic and procedural information. To represent this, we introduce four trainable embedding matrices tailored for diagnoses and procedures in different branches, respectively, denoted as $E_{I_d} \in \mathbb{R}^{|D_I| \times dim}$, $E_{I_p} \in \mathbb{R}^{|P_I| \times dim}$, and $E_{C_d} \in \mathbb{R}^{|D_C| \times dim}$, $E_{C_p} \in \mathbb{R}^{|P_C| \times dim}$, where dim denotes the dimensionality of the embeddings, $|D_I|$, $|P_I|$, $|D_C|$ and $|P_C|$ represent the total number of diagnosis and procedure codes for ICD and CCS code branches, respectively. Each row within E_{I_d}, E_{I_p}, E_{C_d}, and E_{C_p} retain a vectorial representation for a unique diagnosis or procedure. At the tth visit, given the multi-hot encoded vectors for diagnoses and procedures, we extract the pertinent embeddings and aggregate them through an embedding operator $Emb(\cdot,\cdot)$:

$$d_{I_e}^t = Emb(d_I^t, E_{I_d}); p_{I_e}^t = Emb(p_I^t, E_{I_p}), \tag{1}$$

$$d_{C_e}^t = Emb(d_C^t, E_{C_d}); p_{C_e}^t = Emb(p_C^t, E_{I_p}), \tag{2}$$

where $d_{I_e}^t \in \mathbb{R}^{S_{I_d} \times dim}$, $p_{I_e}^t \in \mathbb{R}^{S_{I_p} \times dim}$, $d_{C_e}^t \in \mathbb{R}^{S_{C_d} \times dim}$, $p_{C_e}^t \in \mathbb{R}^{S_{C_p} \times dim}$, S_{I_d}, S_{I_p}, S_{C_d} and S_{C_p} are the sequence length for diagnosis and procedures of two branches.

4.3. Knowledge enhanced patient representation

In this subsection, we utilize the Transformer Encoder (Vaswani et al., 2017) (TE) to model each sequence representation, capturing the long-term dependency of different medical codes. In addition, we innovatively construct multiple medical knowledge graphs and design a graph knowledge enhancement (GKE) module to integrate global relationship information from various code branches for patient representation enhancement. The GKE module aims to integrate the relationship information of the same code across branches from a global perspective to alleviate the problem of imbalanced code distribution. Through TE module and sum the output sequence matrices into vectors, we obtain the feature representations of the above four sequences: $d^t_{I_r} \in \mathbb{R}^{dim}$, $p^t_{I_r} \in \mathbb{R}^{dim}$, $d^t_{C_r} \in \mathbb{R}^{dim}$, $p^t_{C_r} \in \mathbb{R}^{dim}$. Because the four code sequences mentioned above use TE encoding, we explain the TE calculation process for one sequence to simplify the description. Next, we provide additional explanations on the calculation process of TE in detail.

The Transformer Encoder contains a multi-head self-attention mechanism, it can be used to focus on the medical code that is most important for the final output in each input sequence. Specifically, we first generate query, key, and value matrices I_q, I_k, and I_v. To simplify the description, we consider every sequence embedding representation as the TE's input I. The calculation process of each self-attention head is denoted as $Att(\cdot)$.

$$I_q, I_k, I_v = IW^q, IW^k, IW^v, \tag{3}$$

$$Att(I_q, I_k, I_v) = \text{Softmax}(\frac{I_q \times I_k^T}{\sqrt{d}})I_v, \tag{4}$$

where I_q, I_k, and I_v denote the learnable parameters.

Then, we concat the outputs of each head to obtain multi-head attention representation.

$$I_h = (\text{Att}_{h_1}(\cdot) \| ... \| \text{Att}_{h_n}(\cdot)). \tag{5}$$

where $\|$ denotes the concat operation.

Subsequently, we utilize a feed-forward layer and a layer normalization function LayerNorm(·) to filter the critical features of I_h and get the final feature representation I_o, as follows:

$$I_f = \text{ReLU}(I_h W_1 + b_1)(W_2 + b_2), \tag{6}$$

$$I_o = \text{LayerNorm}(\text{FFN}(I_f) + I), \tag{7}$$

where ReLU(·) and FFN(·) denote an activation function and a feed-forward layer, respectively. W_1 and W_2 are trainable parameter matrices, and b_2 is a bias parameter.

Graph knowledge enhancement: We first capture the global relationships of medical code at the label level and then explore knowledge enhancement at the feature level. At the label level, we introduce a self-built knowledge graph to fuse global code relationships between different code branches, which can alleviate patient representation bias caused by imbalanced medical entity distribution.

The $\mathcal{G}_d$ and $\mathcal{G}_p$ contain the correlation relationship of diagnosis and procedure codes between two branches. Integrating the relationship between two branches at the label level can help alleviate the class imbalance problem of previous medical entities and enhance patient representation. Subsequently, a two-layer Graph Convolutional Network (GCN) (Kipf & Welling, 2017) is employed to effectively capture refined embeddings within the two Knowledge Graphs (KGs). Given the parameter matrices E_d, E_p, along with the $\mathcal{G}_d$ and $\mathcal{G}_p$, we employ the GCN layer to obtain a flexible node-embedding architecture. The calculation formula is as follows:

$$GCN(X, A) = \sigma(D^{-\frac{1}{2}} \widetilde{A} D^{-\frac{1}{2}} XW), \tag{8}$$

where X and A denote feature information of nodes and adjacency matrix, respectively. W is a learnable parameter matrix. $\widetilde{A}$=A+Q, Q represents the identity matrix and D represents degree matrix of $\widetilde{A}$.

$$G^e_d = GCN((GCN(E_d, G_d)), G_d), \tag{9}$$

$$G^e_p = GCN((GCN(E_p, G_p)), G_p), \tag{10}$$

where $G^e_d \in \mathbb{R}^{|D_I| \times dim}$ and $G^e_p \in \mathbb{R}^{|P_I| \times dim}$ are the output of the two GCN layers, G_d and G_p are the adjacency matrices corresponding to $\mathcal{G}_d$ and $\mathcal{G}_p$.

After obtaining the KGs embedding, we inject the diagnosis and procedure representations of the CCS branch d^t_C, p^t_C into the ICD code branch d^t_I, p^t_I at the label level through a graph knowledge enhancement operation, as follows:

$$d^t_{I'} = GU(G^e_d \otimes d^t_{C_r}) \oplus d^t_{I_r}, \tag{11}$$

$$p^t_{I'} = GU(G^e_p \otimes p^t_{C_r}) \oplus p^t_{I_r}, \tag{12}$$

where $GU(\cdot)$ represents a gating unit used to adjust the model's focus on important knowledge information.

Then, we concat the final diagnosis and procedure representations of each code branch to obtain patient representations for different branches, as follows:

$$v^t_I = concat(d^t_{I'}, p^t_{I'}), \tag{13}$$

$$v^t_C = concat(d^t_{C_r}, p^t_{C_r}), \tag{14}$$

where v^t_I and v^t_C denote the final patient representations based ICD and CCS code branches, respectively.

4.4. Drug representation

Previous models typically obtain the final patient representation by modelling longitudinal historical visit information of patients and then interacting with the drug representation to obtain recommended drug combinations. They neglect to search directly for important drugs in the historical drug sets, and insufficient patient representation may make it difficult to capture valuable drug information accurately, introducing unnecessary noise information. Therefore, we design a dual-path drug prediction module to predict drugs from different perspectives at the visit level and drug level, respectively.

At the visit level, we interact between patient and drug representations to explore potential drug features. The process is as follows:

$$M_I^t = v_I^t \otimes M_e, \tag{15}$$

$$M_C^t = v_C^t \otimes M_e, \tag{16}$$

where $M_e \in \mathbb{R}^{|M| \times 2dim}$, M_I^t and M_C^t denote the different code branches' drug prediction scores at the visit level, respectively.

At the drug level, we aim to directly select the critical drugs for this visit from past medication records. We design a Drug-wise Signal Recognizer (DSR) to guide the model in selecting drugs closely related to the current visit at each past visit. M^n denotes the drug sets of past visits and $n \in [1, t-1]$. Finally, by adding up the drugs selected from each past visit, we can obtain the entire set of drugs retrieved from the historical visits. The calculation process of DSR is as follows:

$$M_{I'}^t = \sum_{n=1}^{t-1} [Readout(concat(v_I^t, v_I^n)) \odot M^n], \tag{17}$$

$$M_{C'}^t = \sum_{n=1}^{t-1} [Readout(concat(v_C^t, v_C^n)) \odot M^n], \tag{18}$$

where the $Readout(\cdot)$ is a linear layer to adjust output distribution, and $\odot$ denotes the dot product operation.

Adaptive aggregation. We inject the recommended results from the CCS branch into the ICD drug combination obtained from the ICD branch to enhance the prediction performance. The process is as follows:

$$\hat{y}_C^t = \sigma(M_{C'}^t \oplus M_C^t), \tag{19}$$

$$\hat{y}_I^t = \alpha \times \sigma(M_{I'}^t \oplus M_I^t) \oplus \beta \times \hat{y}_C^t, \tag{20}$$

where $\hat{y}_C^t$ and $\hat{y}_I^t$ respectively represent the drug prediction probabilities of the two branches. α and β are two hyper-parameters.

Finally, we generate the predicted drug set $\hat{o}^t$ through the threshold ϕ:

$$\hat{o}_i^t = \begin{cases} 1, & \hat{y}_i^t > \phi \\ 0, & Otherwise \end{cases}. \tag{21}$$

4.5. Training

Drug recommendation can be seen as a multi-label classification task, and our KERL uses two standard loss functions in the classification task for training: binary cross entropy $\mathcal{L}_{bce}$ and multi-label hinge loss $\mathcal{L}_{multi}$. According to Chen et al. (2023), Yang, Xiao, Ma, et al. (2021), the definitions are as follows:

$$\mathcal{L}_{bce} = -\frac{1}{T} \sum_{t=1}^{T} \sum_{i=1}^{|M|} m_i^t \log(\hat{y}_i^t) + (1 - m_i^t) \log(1 - \hat{y}_i^t), \tag{22}$$

$$\mathcal{L}_{multi} = \frac{1}{T} \sum_{t=1}^{T} \sum_{i,j:m_i^t=1,m_j^t=0} \frac{\max(0, 1 - (\hat{y}_i^t - \hat{y}_j^t))}{|M|}, \tag{23}$$

where $m^t \in \{0,1\}^{|M|}$ denotes the true label set of drug recommendation, $\hat{y}^t$ is the prediction value of drug, T is the visit number of a patient.

We introduce Kullback Leibler (KL) divergence to minimize the difference between the predicted probability distributions of two code branches to accelerate model convergence, as follows:

$$\begin{aligned} \mathcal{L}_D^i &= \frac{1}{T} \frac{1}{n} \sum_{t=1}^{T} \sum_{i=1}^{n} D_{KL} \sigma((\frac{\hat{y}_I^t}{T}) \parallel \sigma(\frac{\hat{y}_C^t}{T})) \times T_e^2 \\ &= \frac{1}{T} \frac{1}{n} \frac{1}{|M|} \sum_{t=1}^{T} \sum_{i=1}^{n} \sum_{s=1}^{|M|} P_{is}(log_{P_{is}} - log_{q_{is}}) \times T_e^2, \end{aligned} \tag{24}$$

where n and T_e denote the number of samples and a temperature parameter, respectively.

Combining Eqs. (22), (23), and (24), the joint loss function of the proposed KERL can be expressed as follows:

$$\mathcal{L}_{joint} = \lambda_1 \cdot \mathcal{L}_{bce} + \lambda_2 \cdot \mathcal{L}_{mlh} + \lambda_3 \cdot \mathcal{L}_D^i, \tag{25}$$

where λ_1, λ_2, and λ_3 represent hyper-parameters controlling the importance of each loss function.

Table 2
Statistics of the MIMIC-III and MIMIC-IV datasets.

Items	MIMIC-III	MIMIC-IV
#Patients	6,350	8,546
#Clinical events	15,032	19,592
#ICD/CCS-Diagnoses	1,958/241	1,892/217
#ICD/CCS-Procedures	1,430/202	4,798/200
#Drugs	131	131
Avg/max #of visits	2.37/29	2.29/28
Avg/max #of diagnoses	13.63/39	17.37/39
Avg/max #of procedures	4.53/32	4.02/34
Avg/max #of drugs	19.19/53	16.74/64

5. Experiments

In this section, we first introduce the experimental settings. Then, we present the main experiments and ablation studies about the main components of our method. Next, we have a further analysis about two core modules. Finally, we conduct the case analysis and interpretability analysis to demonstrate the effectiveness and interpretability of the model KERL.

5.1. Experimental settings

Datasets. We assess our model using the MIMIC-III (Johnson et al., 2016) and MIMIC-IV (Johnson et al., 2020) datasets. The MIMIC-III dataset is a large, freely accessible, and comprehensive research database of intensive care unit (ICU) patient records, spanning from June 2001 to October 2012. The MIMIC-IV dataset, the latest iteration in the MIMIC series, encompasses a lot of clinical data, such as demographics, vital signs, laboratory results, procedures, and medication administration.

Following the data preprocessing of previous work (Li, Zhang, Li, et al., 2023; Wu, Qiu, et al., 2022), we obtain 131 drugs suitable for recommendation, with the ATC (Anatomical Therapeutic Chemical) third level code designated as the target label. Each ATC third level code corresponds to one or more medications. We utilize the top 40 most prevalent drug-drug interactions (DDIs) listed in the TWOSIDES dataset (Tatonetti et al., 2012). According to the existing work (Wu, Qiu, et al., 2022; Yang, Xiao, Ma, et al., 2021), we divide the dataset into training, validation, and test sets at a 4:1:1 rate.

CCS[1] (Clinical Classification Software) is a medical classification system primarily used to classify and code diseases for better management and statistical purposes of medical expenses. It is maintained by the Healthcare Cost and Utilization Project (HCUP) and offers medical practitioners, insurance entities, and researchers a user-friendly coding methodology for diagnoses and procedures. We convert the diagnosis and procedure codes in EHRs into coarse-grained CCS codes for training on both code branches. For example, in the MIMIC-III dataset, 1958 diagnostic codes are converted into 241 CCS codes, and 1430 procedure codes are converted into 202 CCS codes. Table 2 shows the detailed data statistics of both datasets.

Implementation details. All experiment results are reported using an NVIDIA 3080Ti GPU to train. For the two datasets, the embedding dimension size and learning rate are set at 64 and 1×10^{-4}, respectively. Moreover, we utilize the Adam (Kingma & Ba, 2015) optimizer to optimize model parameters and run within 50 epochs. We report the mean and standard deviation from 10 rounds as the final results. In the knowledge enhanced patient representation module, we employ two-layer GCN to encode node embedding in knowledge graphs. For drug prediction, the predefined threshold ϕ is set to 0.35 to select which drug is recommended. In Kullback Leibler loss, the temperature parameter T_e is set as 4. Moreover, we set λ_1, λ_2, and λ_3 as 0.5, 0.01, and 1, to control the importance of each loss function.

Evaluation metrics. We evaluate performance using four metrics, Jaccard similarity (Jaccard), Precision-Recall AUC (PRAUC) and F1-score (F1) and Drug-Drug Interaction (DDI) rate. The Jaccard, PRAUC, and F1 primarily focus on assessing the accuracy of the recommendations, in line with standard practices in the broader domain of recommendation systems. The DDI rate is used to calculate the adverse interaction between two drugs, and the lower the metric, the better. The calculation methods for different metric formulas are as follows:

F1. The F1 score is a widely used metric for evaluating a model's performance. It is derived from the harmonic mean of precision and recall, providing a balanced perspective that accounts for the proportion of true positive predictions among all positive predictions (precision) and the proportion of true positive predictions among all actual positives (recall). This harmonic balance ensures that the F1 score offers a comprehensive assessment of the model's ability to make accurate predictions while minimizing different types of prediction errors, as follows:

$$Pre^t = \frac{1}{T}\frac{|\{i|m_i^t = 1\} \cap \{j|\hat{o}_i^t = 1\}|}{|\{j|\hat{o}_i^t = 1\}|}, \tag{26}$$

$$Rec^t = \frac{1}{T}\frac{|\{i|m_i^t = 1\} \cap \{j|\hat{o}_i^t = 1\}|}{|\{i|m_i^t = 1\}|}, \tag{27}$$

[1] https://www.who.int/standards/classifications/classification-of-diseases

$$F1 = \frac{1}{N}\sum_{t=1}^{T}\sum_{i}^{N}\frac{2\times Pre^t \times Rec^t}{Pre^t + Rec^t}, \tag{28}$$

where $\hat{o}_i^t$ denotes the predicted drug label vector, m_i^t denotes the ground truth drug label vector, N denotes the number of patients, T is the visit number of a patient.

PRAUC. PRAUC, or Precision-Recall Area Under Curve, is a metric employed to assess the performance of a model in scenarios where the positive class is of particular interest and imbalance between classes is present. It measures the trade-off between precision and recall across different threshold settings by calculating the area under the precision–recall curve. By summarizing this trade-off into a single score, PRAUC comprehensively evaluates the model's effectiveness in handling imbalanced datasets, mainly when focusing on optimizing precision and recall. The calculation formula is as follows:

$$PRAUC = \frac{1}{T}\frac{1}{N}\sum_{t=1}^{T}\sum_{i}^{|M|-1}(Pre_{i+1}^t - Pre_i^t) * Rec_i^t, \tag{29}$$

where Pre_i^t and Rec_i^t denote the precision and recall rate with the prediction probability of the drug with the i ranking probability in the tth visit as the threshold value, T is the same as those in the definition of F1.

Jaccard. To validate the model's fitting ability at the set level, we utilize the Jaccard metric as a consistent measure of similarity between the recommended drug set and the true drug set, as follows:

$$Jaccard = \frac{1}{T}\frac{1}{N}\sum_{t=1}^{T}\sum_{i}^{N}\frac{|\{i|m_i^t = 1\} \cap \{j|\hat{o}_i^t = 1\}|}{|\{i|m_i^t = 1\} \cup \{j|\hat{o}_i^t = 1\}|}, \tag{30}$$

where the meanings of $\hat{o}_i^t$, m_i^t, T and N are the same as those in the definition of F1.

The DDI rate represents the proportion of drug pairs with adverse interactions in the model's predicted results. In the adjacency matrix A_e, $(A_e)_{i,j} = 1$ indicates the presence of adverse interactions between the ith drug and the jth drug. We use A_e to calculate the DDI rate of the model's predicted results $\hat{o}_i^t$.

$$DDI = \frac{1}{T}\frac{1}{N}\sum_{t=1}^{T}\sum_{t}^{N}\frac{\{i,j|(A_e)_{i,j} = 1, \hat{o}_i^t = 1, \hat{o}_j^t = 1\}}{|M|^2}, \tag{31}$$

Baselines. To evaluate our KERL, we select several popular drug recommendation models for comparison: ECC (Read et al., 2011), RETAIN (Choi et al., 2016), LEAP (Zhang et al., 2017), GAMENet (Shang, Xiao, et al., 2019), SafeDrug (Yang, Xiao, Ma, et al., 2021), MICRON (Yang, Xiao, Glass, & Sun, 2021), COGNet (Wu, Qiu, et al., 2022), DGCL (Li, Zhang, Li, et al., 2023), and MoleRec (Yang et al., 2023). The description of baselines are shown in the supplementary materials.

We compare our model with multiple baselines, as follow:

- ECC (Read et al., 2011) denotes a classifier chains method for multi-label classification.
- RETAIN (Choi et al., 2016) designs reverse time attention model to enhance the model's interpretability.
- LEAP (Zhang et al., 2017) utilizes a recurrent decoder to capture the label dependency relationship and uses a content-based attention mechanism to extract drug-disease mapping.
- GAMENet (Shang, Xiao, et al., 2019) is a graph-augmented memory network, which utilizes GCN to integrate multiple knowledge graphs into memory components for generating safe and effective recommendation combinations.
- SafeDrug (Yang, Xiao, Ma, et al., 2021) is a safe and effective drug recommendation model by controlling DDI loss, which introduces drug molecular structure to model DDI explicitly.
- MICRON (Yang, Xiao, Glass, & Sun, 2021) designs a residual-based inference framework that develops an efficient sequential updating mechanism based only on the recent visit's new diagnoses or procedure features.
- COGNet (Wu, Qiu, et al., 2022) develops a copy-or-predict mechanism to leverage historical recommendations.
- DGCL (Li, Zhang, Li, et al., 2023) designs a distance detection loss to calculate the similarity between the current case and past visits and employs graph contrastive learning to reduce drug-drug interactions.
- MoleRec (Yang et al., 2023) constructs a hierarchical architecture to capture molecular substructures that contribute to improving treatment efficacy and recommendation safety.

5.2. Results and analysis

To verify the superiority of the proposed model KERL, we compare 9 competitive models on the MIMIC-III and MIMIC-IV datasets. We leverage Jaccard, F1 and PRAUC to assess all models' performance. Table 3 shows that KERL exceeds all baselines on major indicators. Our model achieves improvements of 2.15%, 2.09%, 1.92% and 2.09%, 2.74%, 1.96% in F1-score, PRAUC and Jaccard, compared to the most competitive performance.

Specifically, our model demonstrates significant superiority in major indicators compared to the methods (ECC and LEAP) based solely on current visits. The instance-based methods only focus on medical information in a single visit and ignore capturing patient history information from a global perspective, thus exhibiting limited recommendation performance. The models (RETAIN, GAMENet, MICRON, COGNet, DGCL) integrate past medical information to obtain a comprehensive representation of patients, thereby enhancing model performance. However, RETAIN, GAMENet and DGCL do not directly model the patient's past drug

Table 3
Performance comparison on the MIMIC-III/MIMIC-IV. Numbers in bold and underlined indicate the current and previous best performance, respectively. For Jaccard, PRAUC and F1-score, the higher the better, while for DDI rate and Avg drugs, the lower the better.

Methods	Jaccard ↑	PRAUC ↑	F1-score ↑	DDI rate ↓	Avg medications
ECC	0.4996/0.4233	0.6844/0.7284	0.6569/0.5680	0.0846/0.0771	18.0722/8.1070
LEAP	0.4521/0.4287	0.6549/0.5506	0.6138/0.5820	0.0731/0.0592	18.7138/11.5198
RETAIN	0.4887/0.4239	0.7556/0.6798	0.6481/0.5791	0.0835/0.0939	20.4051/10.8602
GAMENet	0.5237/0.4963	0.7775/0.7508	0.6783/0.6514	0.0861/0.0890	27.2145/18.4426
SafeDrug	0.5233/0.5000	0.7742/0.7485	0.6764/0.6557	**0.0615**/0.0575	19.9178/14.4705
MICRON	0.5260/0.4921	0.7773/0.7514	0.6790/0.6482	0.0641/**0.0574**	18.9800/16.8439
COGNet	0.5289/0.5081	0.7675/0.7547	0.6829/0.6627	0.0831/0.0837	28.0900/24.6800
DGCL	0.5255/0.4993	0.7738/0.7535	0.6801/0.6542	0.0836/0.0735	28.6253/16.6284
MoleRec	0.5271/0.4930	0.7717/0.7426	0.6816/0.6503	0.0726/0.0961	21.6489/18.7085
KERL	**0.5504***/**0.5290***	**0.7982***/**0.7821***	**0.7021***/**0.6823***	0.0645*/0.0612*	19.7813/17.1765

* Denotes that the result is statistically significant when compared with the second best with p-value < 0.05.

Table 4
Ablation study for different components of KERL on MIMIC-III and MIMIC-IV. Numbers in bold signify the best performance.

	MIMIC-III			MIMIC-IV		
Methods	F1-score	PRAUC	Jaccard	F1-score	PRAUC	Jaccard
KERL w/o D	0.6703 ± 0.0017	0.7715 ± 0.0020	0.5136 ± 0.0012	0.6504 ± 0.0011	0.7511 ± 0.0014	0.4940 ± 0.0011
KERL w/o P	0.6842 ± 0.0014	0.7834 ± 0.0016	0.5287 ± 0.0010	0.6598 ± 0.0010	0.7604 ± 0.0015	0.5014 ± 0.0010
KERL w/o TE	0.6877 ± 0.0015	0.7874 ± 0.0015	0.5257 ± 0.0011	0.6703 ± 0.0011	0.7704 ± 0.0014	0.5094 ± 0.0013
KERL w/o KE	0.6874 ± 0.0013	0.7868 ± 0.0016	0.5341 ± 0.0010	0.6683 ± 0.0011	0.7697 ± 0.0013	0.5114 ± 0.0011
KERL w/o DDP	0.6893 ± 0.0011	0.7903 ± 0.0013	0.5359 ± 0.0010	0.6695 ± 0.0011	0.7704 ± 0.0012	0.5127 ± 0.0009
KERL	**0.7021 ± 0.0011**	**0.7982 ± 0.0013**	**0.5504 ± 0.0009**	**0.6823 ± 0.0010**	**0.7821 ± 0.0012**	**0.5290 ± 0.0008**

history, which shows restrictive performance compared to our model. Moreover, MIRCON has been observed to solely identify valuable drugs from a single recent visit, thereby rendering the modelling of a patient's comprehensive medication history incomplete and inadequate. In contrast, COGNet introduces a hierarchical selection mechanism to enhance performance, which is designed to comprehensively recreate useful drugs from an entire patient's historical records. However, this replication process is characterized by its coarse granularity, which may result in the model predicting redundant drugs. While this approach boosts model performance, it unfortunately overlooks safety considerations, ultimately contributing to a heightened incidence of drug-drug interactions (DDIs).

SafeDrug and MoleRec introduce additional medical knowledge on the basis of integrating longitudinal patient information to enhance representation learning and improve model recommendation performance. Among them, SafeDrug introduces drug molecular structure knowledge to model DDI explicitly and achieves the lower DDI rate. MoleRec develops a molecular substructure-aware encoding network to capture the association between drug substructures and the target disease. The existing methods struggle to maintain high security while pursuing high performance. Additionally, KERL achieves a lower DDI rate than most methods, which indicates the system's capability to effectively balance safety and accuracy.

5.3. Ablation study

We perform an ablation study to illustrate the efficacy of the core components of KERL. We utilize Jaccard, F1 and PRAUC to assess all models' performance based on the MIMIC-III and MIMIC-IV datasets. This study includes 5 variants:

- **KERL w/o D** removes the diagnosis information, utilizing only the procedure information for patient representation.
- **KERL w/o P** excludes the procedure information, relying solely on the diagnosis information for patient representation.
- **KERL w/o TE** removes the Transformer Encoder, do not capture the long-term dependency of different medical codes in various sequences.
- **KERL w/o KE** does not have a knowledge enhancement to obtain patient representation, utilizing only the ICD code sequence as data input for drug recommendation.
- **KERL w/o DDP** removes drug-wise signal recognizer without modelling the patient's past drug history.

Table 4 shows the performance for the multiple variants of KERL. As expected, when medical information (i.e. diagnosis or procedure sequence) is lost, accurately representing the patient's health status is difficult, and performance is poor. When we do not use TE to model the sequence representation of patients, performance decreases significantly. This means that capturing the long-term dependencies of entities in the sequence is important for improving representation learning. When we remove the KE module and only use the ICD code branch for the final prediction, performance decreases significantly, indicating that integrating CCS knowledge can enhance feature representation to improve the prediction accuracy. When DDP is removed, performance decreases, indicating that the past drug history contains crucial information for the next visit, and our designed DDP can effectively extract valuable historical drug information. The proposed KERL demonstrates superior performance, with results that are statistically significant compared to the second-best model (p-value < 0.05). Overall, the complete KERL outperforms all ablation models, which means each core component of our model is essential.

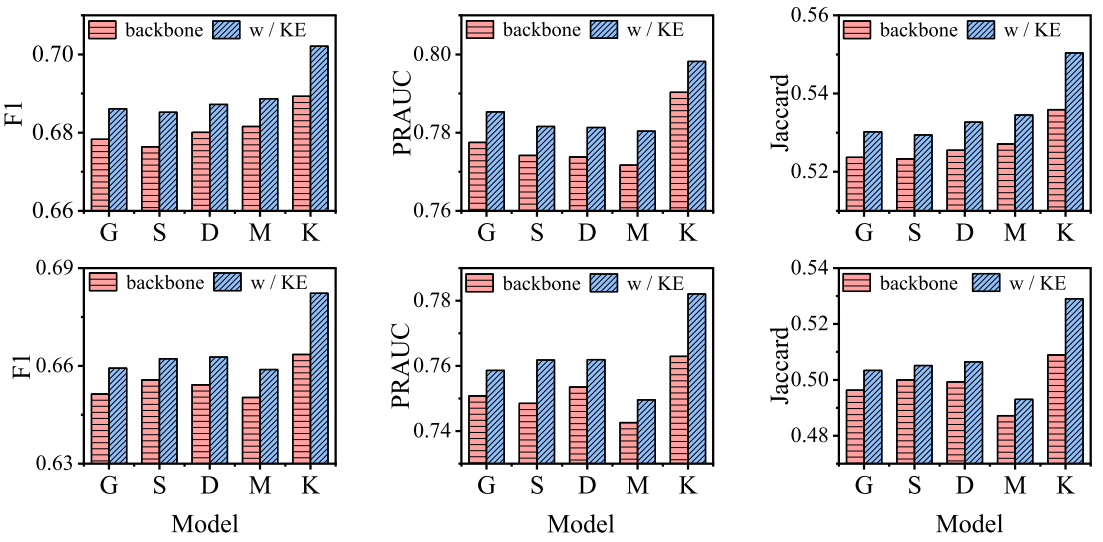


Fig. 5. Results of MIMIC-III and MIMIC-IV when trained with/without the KE module, corresponding to the first and second rows, respectively.

5.4. Effect of knowledge enhanced patient representation

To further demonstrate the effectiveness of the Knowledge Enhanced patient representation (KE) module, we select four replicable and competitive models: GAMENet (G), SafeDrug (S), DGCL (D), MoleRec (M) and KERL (K), and then add the KE module to these four backbones. We utilize Jaccard, F1 and PRAUC to assess all models' performance based on the MIMIC-III and MIMIC-IV datasets.

As shown in Fig. 5, after adding the proposed KE strategy, which integrates CCS knowledge to enhance representation learning, the performance of all the backbone models is improved, and the F1 score is improved by 0.87% and 1.03% on average based on both datasets. Meanwhile, our proposed KERL model shows the best performance. The observed performance improvements can be ascribed to the KE's capacity to incorporate external knowledge, which converts input sequences into different code branches for knowledge distillation, sharing relationship information among code branches. This process enhances patient representation by alleviating the imbalanced distribution in the input code sequences, thereby significantly improving the accuracy of drug recommendations across various models. In summary, our knowledge enhancement strategy improves the performance of all metrics across multiple models, demonstrating the generalization and effectiveness of our method.

5.5. Effect of dual-path drug prediction

To demonstrate that the DDP module can better integrate past drug information, we select four longitudinal information-based models (GAMENet, SafeDrug, DGCL, and MoleRec) for comparative experiments on two datasets. Due to the small number of samples with more than five visits, we select samples from the first five visits and divide the dataset into five parts for separate experiments. We utilize Jaccard, F1 and PRAUC to assess all models' performance based on the MIMIC-III and MIMIC-IV datasets. The final experimental results reported in Table 3 are derived from the weighted sum of these five parts.

Fig. 6 visually shows that the performance of KERL continued to improve in the first three visits of the experiment, and the predictive performance of more than one visit is higher than that of only one visit, indicating that KERL can fully utilize historical drug information. In contrast, the performance of the other models is lower than that of KERL at all visits, and as the number of visits increases, the performance improvement is not significant or even partially decreases. In particular, in experiments involving five visits, the four models' performance sharply decreases on the MIMIC-IV dataset. These models typically construct comprehensive patient representations by integrating historical visit information and interacting with drug representations. Their performance decline may be attributed to the integration of unrelated visits that may introduce noise to the current patient representation and insufficient patient drug interaction, causing unnecessary information loss.

5.6. Interpretability analysis

To demonstrate how KERL selects drugs from past drugs to enhance recommendation performance, we randomly select a patient sample and visualize the process on the MIMIC-III dataset. This patient has undergone a total of 5 visits, and we treat the last visit as the current visit to explore the impact of the drugs recommended in the previous 4 visits on the current recommendation. The first row denotes the drugs representing the ATC third level code, and the vertical axis represents the patient's tth visit and the weight of the drug's contribution to the current visit. Green and red respectively represent the positive and negative drug effects of historical drugs for current predicted drugs. The positive effect is manifested by the high contribution of historical drugs to the true

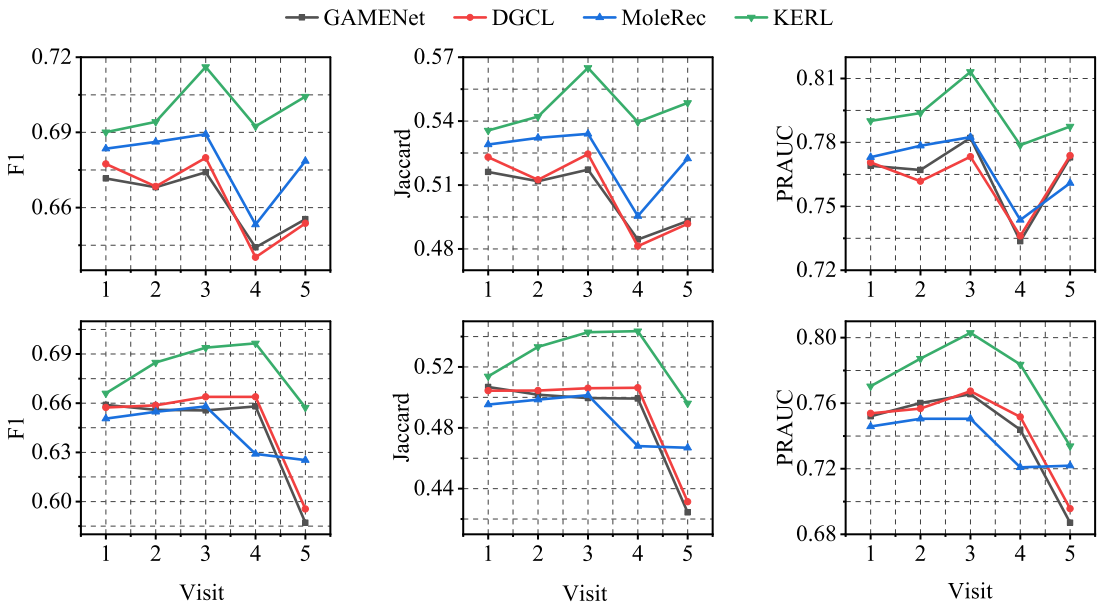


Fig. 6. Effects of the number of history visits for different models based on the MIMIC-III and MIMIC-IV, corresponding to the first and second rows, respectively.

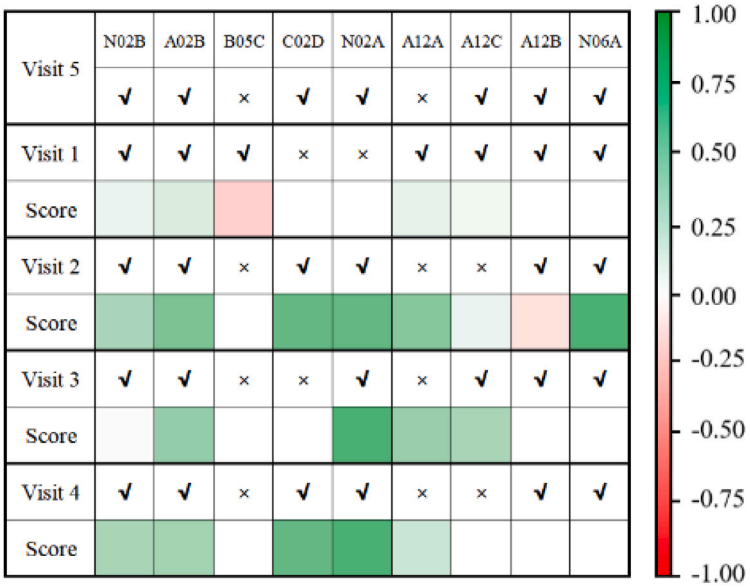


Fig. 7. Visualizations of drug selection from historical drugs. Green and red respectively represent the positive and negative drug effects of historical drugs for current predicted drugs, with darker colours indicating a greater degree of correlation.

value of the current medication, where historical drugs contribute to the recommendation of current drugs. The negative effect is manifested as the absence of a particular drug in the actual value of the current visit, but the drug has been used in the historical drug history. The model can recognize that the historical record of the drug has a negative effect on the recommendation of the current drug, so the weight generated by the model is negative.

As shown in Fig. 7, we visualize the effect scores of past drugs on the current drugs, and it can be seen that for the upcoming predicted drugs, like N02B, A02B, C02D, N02A, A12A, A12C and N06A, the model successfully focuses on reusable drugs in the previous drug sets. Moreover, the model can also reasonably identify the misleading effects of past drugs on the current situation, such as drug B05C appearing on the first visit rather than not appearing on the prediction set and drug A12B not appearing on the current visit rather than appearing on the previous prediction sets. This visualization result demonstrates that our model can

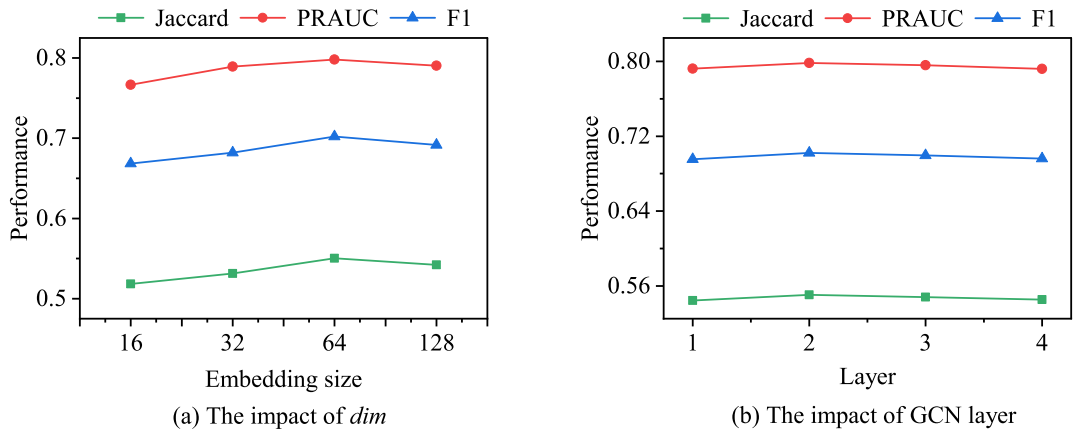


Fig. 8. Hyperparameter impact. Subfigure (a)–(b) shows the influence of different embedding dimension dim and GCN layers for KERL on MIMIC-III dataset.

effectively mine valuable and reusable drugs from past drugs. Furthermore, it provides satisfactory transparency and enhances the model's interpretability.

5.7. Hyperparameter study

In this subsection, we detail the hyperparameter selection process for several important hyperparameters within our model, specifically for the embedding dimension (dim) of E_{I_d}, E_{I_p}, and E_{C_d}, E_{C_p} within the set [32, 64, 128], and the layer of GCN within the range [1, 2, 3]. We report Jaccard, F1 and PRAUC to assess models performance based on the MIMIC-III dataset.

As illustrated in Fig. 8(a), we conduct an extensive series of scaling experiments involving adjustments to the embedding size to evaluate the impact of the embedding dimension (dim) on the efficacy of our proposed framework, KERL. The experimental results indicate that the performance of KERL exhibits an upward trend as the embedding dimension increases, reaching an optimal point. Specifically, a peak in performance is noted at an embedding dimension of 64. However, beyond this point, further augmentation in the embedding dimension resulted in a decline in the model's performance. This decrement in performance may be attributed to two primary factors: an increased susceptibility to noise within the framework and a heightened vulnerability to overfitting, both of which are associated with excessively high dimensionality. Therefore, we set the optimal embedding dimension for our framework as 64.

As depicted in Fig. 8(b), we thoroughly analyse the parameter sensitivity related to the number of Graph Convolutional Network (GCN) layers, varying this parameter within the range of 1, 2, and 3. The objective is to ascertain the influence of the GCN layer count on the performance of our proposed model. The results reveal a distinct performance trend as the number of GCN layers increases. Notably, the model achieves optimal performance when configured with 2-layer GCN. Below this optimal configuration, with only 1-layer GCN, the model's capacity to capture complex graph structures may be limited, leading to suboptimal performance. Conversely, increasing the number of GCN layers beyond 2 to 3 decreases performance. This decline is attributed to issues such as overfitting and vanishing gradients, which are common challenges in deep neural networks with excessive layers. Based on these comprehensive findings, our model's optimal number of GCN layers is set as 2.

5.8. Case study

As shown in Table 5, we conduct a case study on one patient selected randomly from the test set of a real-world MIMIC-III dataset. It can be observed that this patient has a total of three medical visit records. We use ICD-9 codes for diagnosis representation and convert drug NDC codes into ATC-third codes to represent the final recommended drugs. The "correct" represents the set of drugs correctly predicted in the true data set. The "missed" means the drugs ignored in the predicted drug set. Conversely, the "unseen" enumerates medication combinations that the models forecast but are absent from the ground-truth dataset. The patient is diagnosed with the following conditions during the first visit: difficulty breathing, accompanied by high blood pressure, urinary system disorders, and bacterial infection. In the second diagnosis, sepsis caused by pseudomonas aeruginosa, urinary tract infection, kidney stones, and postoperative infection bleeding are identified as symptoms. The third diagnosis includes kidney stones, urinary system disorders, arrhythmia, and multiple sclerosis.

To further investigate how our model performs in recommending therapeutic drugs for patients, we compare KERL with the most recent and competitive baseline, MoleRec. As the patient's condition progresses, it can be observed that in the complex medical environment with multiple visits, our model can still predict most of the drugs in the actual prescription and even recommends some drugs not present in the authentic prescription. In contrast, MoleRec shows a significant drop in accuracy and experiences substantial performance changes in recommending drugs. It is worth noting that our model accurately predicts the majority of drugs within actual prescriptions and also suggests some drugs that are not present in the authentic prescriptions. Specifically, in the second

Table 5
Example recommended drugs for a patient with three visits. C, M, and U denote correct, missed and unseen drug codes, respectively. The underline indicates that the predicted drugs do not appear in the true drug set.

Visits	Diagnoses	True Prescription & Total	KERL & C/M/U	MolRec & C/M/U
1	'4019', '2875', '6930', '78060', '5990', '04112', '34830', 'V4986', 'E9308', '3453', '78609', '0417', 'E9394', 'V1302', '340', '78065', '78603'	'N02B', 'A01A', 'A02B', 'A06A', 'B05C', 'A12C', 'A07A', 'B01A', 'J01D', 'N03A', 'C08C', 'R01A', 'G04B', 'D04A', 'D06B', 'M03B' & 16	'N02B', 'A01A', 'A02B', 'A06A', 'B05C', 'A12C', 'A07A', 'C07A', 'A12B', 'N02A', 'A02A', 'J01M', 'B01A', 'J01D', 'N03A', 'N05A' & 10/6/6	'N02B', 'A01A', 'A02B', 'A06A', 'A12C', 'A07A', 'C03C', 'A12B', 'N05B', 'A02A', 'J01M', 'B01A', 'N03A', 'N05A', 'A04A' & 8/8/7
2	'99859', 'E8788', '99811', '99591', '5990', '2869', '27650', '5920', '04185', '03843', '59654', '340'	'N02B', 'A02B', 'A06A', 'B05C', 'A12A', 'A12C', 'A07A', 'A12B', 'B01A', 'J01D', 'J01G', 'M03B' & 12	'N02B', 'A01A', 'A02B', 'A06A', 'B05C', 'A12A', 'A12C', 'A07A', 'A12B', 'N02A', 'J01M', 'B01A', 'J01D', 'N03A' & 10/2/4	'N02B', 'A01A', 'A02B', 'A06A', 'B05C', 'A12C', 'A07B', 'A12B', 'N02A', 'J01M', 'N03A', 'N06A', 'A09A' & 6/6/6
3	'42789', '5920', 'V1302', '59654', '340'	'N02B', 'A02B', 'A06A', 'A12A', 'A12C', 'A07A', 'A12B', 'N02A', 'B01A', 'J01D', 'A04A', 'G04B', 'M03B' & 13	'N02B', 'A01A', 'A02B', 'A06A', 'B05C', 'A12A', 'A12C', 'A07B', 'A12B', 'N02A', 'B01A', 'J01D', 'N03A', 'A04A' & 10/3/4	'N02B', 'A01A', 'A02B', 'A06A', 'B05C', 'A12A', 'A07A', 'A12B', 'N02A', 'B01A', 'J01M', 'N03A', 'A09A' & 8/5/5

drug prediction, the model recommends N02A, which belongs to the opioid class of drugs, because the patient is diagnosed with postoperative bleeding and infection symptoms during the second diagnosis. This indicates that the model can recommend painkillers for the patient based on their medical history. Furthermore, the patient is diagnosed with multiple sclerosis in the last two diagnoses, which is a neurological disorder, and our model also predicts N03A in both cases. The drug N03A belongs to antiepileptic drugs and can be used for the treatment of multiple sclerosis. Overall, KERL predicts more correct drugs and recommends fewer additional drugs compared to MoleRec, which further indicates the effectiveness of our proposed model.

6. Conclusion

In this paper, we propose a Knowledge Enhanced Representation Learning framework (KERL) for drug recommendation. KERL integrates external knowledge and conducts knowledge distillation to enhance patient representation, which alleviates the data sparsity of low-frequency codes by jointly learning disease codes of different granularities. Meanwhile, it designs a graph knowledge enhancement mechanism to integrate global medical information to alleviate the imbalanced representation of medical information. Moreover, a dual-path drug representation network is developed to model longitudinal historical information from the visit-level and drug-level perspectives. The experiment results on the publicly real-world MIMIC-III and MIMIC-IV datasets show the KERL outperforms existing drug recommendation methods and achieves SOTA performance. Notably, KERL achieves improvements of 2.15%, 2.09%, 1.92% and 2.09%, 2.74%, 1.96% in F1-score, PRAUC, and Jaccard, respectively, compared with state-of-the-art performance. Case analysis and interpretability analysis further demonstrate the effectiveness and interpretability of the model.

7. Limitations and future work

In this section, we delve into the inherent limitations of our current study and highlight potential directions for future research, aiming to advance the field of combination drug recommendation.

Class Imbalance Problem. The challenge of class imbalance poses a significant obstacle in the realm of medication recommendation tasks, particularly evident in the class distribution where a prominent long tail of infrequently occurring diagnoses and procedures is observed. This imbalance in data distribution causes the model to exhibit a bias towards high-frequency samples during the training phase, ultimately leading to suboptimal performance and a lack of generalization to less common scenarios. The class imbalance issue has received relatively scant attention in existing research, highlighting the urgent need for further exploration and innovative solutions in future endeavours. Addressing this imbalance is crucial for developing robust and reliable drug recommendation systems that cater to diverse patient needs and conditions.

Inductive Learning for New Drugs. The limitations posed by the transductive learning assumption in medication recommendation systems present a significant challenge (Yang et al., 2023). Our current model assumes that all drugs and patients appearing in the testing set have been exposed to the model during the training stage, implying a shared drug and patient space between training and testing data. This setting aligns with the literature on transductive learning. However, real-world recommendation systems must interact with an ever-evolving landscape where new, unseen drugs and patients can emerge at any time (Wu et al., 2021). In the context of medication recommendation, this could mean introducing novel drugs or new patients seeking medical care. Ideally, a medication recommendation model should be able to handle these new entities — unseen during training — without requiring re-training or fine-tuning on the new data. This problem set is called inductive learning (Hamilton et al., 2017), which entails the ability of the model to generalize to new entities that were not observed during training. As such, future work in this area must focus on developing models that can effectively address the limitations of transductive learning and embrace the challenges of inductive learning, ensuring that medication recommendation systems remain adaptable and robust in the face of new drugs and patients.

Model Interpretability. Although the proposed knowledge enhanced representation learning network holds the potential for improving existing medication recommendation models, our current focus is primarily on bolstering feature representation by incorporating multiple tasks. However, this approach neglects the intricate interconnections and operational mechanisms among distinct code branches, limiting the model's interpretability. Furthermore, the prevalent medication recommendation models, which predominantly leverage deep learning techniques, retain a significant degree of opacity, resembling a black box in its operational clarity. Consequently, future research endeavours should prioritize advancements in the transparency and interpretability of model recommendations (Ahmed & Alpkoçak, 2022). By doing so, we aim to clarify how the model operates and makes decisions, thereby fostering greater trust and reliability in the medication recommendation process.

Privacy and Security. The constraints posed by privacy and security considerations represent a formidable challenge in medication recommendation. These issues are particularly acute given the sensitive nature of patient data and the potential for misuse or unauthorized access (Ji et al., 2024). Due to these privacy and security concerns, the development of medication recommendation systems faces significant limitations, often necessitating stringent data protection measures and compliance with rigorous regulatory frameworks. However, despite their critical importance, existing research has devoted relatively limited attention to these privacy and security challenges, highlighting the pressing need for further exploration and innovative solutions in future endeavours. Addressing these challenges is essential for fostering trust in medication recommendation systems and ensuring patient data remains secure and confidential.

CRediT authorship contribution statement

Xiaobo Li: Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Xiaodi Hou:** Visualization, Validation, Methodology. **Fanjun Meng:** Validation, Methodology, Data curation, Conceptualization. **Xiaokun Zhang:** Writing – review & editing, Visualization, Validation. **Mingyu Lu:** Project administration, Investigation, Formal analysis. **Hongfei Lin:** Project administration, Investigation, Formal analysis. **Yijia Zhang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by grant from the Natural Science Foundation of China (No. 62072070).

Data availability

Data will be made available on request.

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